

DFT: relaxed primitive cell



Choose a supercell
a conventional cell is often a good start



alm MODE = suggest (**NORDER = 1**)
harmonic displacement patterns



displace.py --pattern_file
small displacements, about 0.01 Å



DFT forces → extract.py → DFSET



alm MODE = optimize (**NORDER = 1**)
LMODEL = ols



harmonic.h5
harmonic force constants (FC2)



Anharmonic
IFCs
needed?

no

anphon
phonons, DOS, thermodynamics



yes
(kappa, SCPH, QHA, ...)

anphon MODE = phonons (**KPMODE = 0**)
FCSFILE = harmonic.h5 → PREFIX.evec



displace.py
--random_normalcoord
--evec PREFIX.evec --temp T
random displacements sampled at temperature T
(or --random on MD snapshots)



DFT forces → extract.py → DFSET



alm MODE = optimize (**NORDER = 2, 3, ...**)
LMODEL = enet or adaptive-lasso
FC2FIX = harmonic.h5



PREFIX.h5
harmonic and anharmonic force constants